Alexandria University

Faculty of Computer and Data Science

Survey Methodology

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Proposal about:

Food Preferences Survey

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Content

Introduction:

Food preferences are an essential aspect of people's daily lives, impacting their health, social interactions, and cultural experiences. As a result, understanding people's food preferences is crucial for improving food services and marketing strategies. The purpose of this proposal is to outline a survey on food preferences and its significance in gathering relevant data for future decision-making processes.

Objectives:

The primary objective of the 'food preferences' survey is to gather data on people's food preferences in terms of their favorite cuisines, ingredients, and dietary restrictions. The survey aims to determine the most preferred food types and explore how they vary from one person to another.

The survey provides valuable insights into people's food preferences, which is expected to aid in developing more efficient and more targeted marketing strategies. Additionally, the data collected can be used to develop new products or menu items that cater to people's preferences and dietary needs. Furthermore, understanding food preferences can help in promoting healthy eating habits and can help reduce food waste.

Participants:

The survey is to be conducted on a number of students from the Faculty of Computer and Data Science, Alexandria University, ranging from 350 to 400 students using an anonymous online platform, which is accessible to a large number of participants. An invitation will be sent to the study groups of FCDS explaining the aim of the survey and how to participate in it. The message and the introduction to the online questionnaire will explain the aim of the survey and provide the guarantee of anonymity of the investigation.

Survey Design:

The survey will be designed in the form of an online questionnaire of 20-30 questions. These questions will provide the following information :

1. **dietary preferences and restrictions (vegetarian, vegan, gluten-free, else)**
2. **food choices and preferences - favorite cuisine, preferred ingredients, frequency of eating out.**
3. **food consumption habits - cooking at home, ordering online, dining out.**
4. **Demographic questions - age, gender, ethnicity, income, and education.**

The questionnaire will be designed to be easy to complete and take no more than 10-15 minutes to finish. It will be structured into sections, making it easier for participants to fill in. Open-ended questions will be kept to a minimum to simplify the data analysis process. Multiple-choice questions will be used for most of the questions, with the option for participants to select more than one answer if applicable.

Data Analysis:

The data collected from the survey will be analyzed using statistical softwares. Powerful statistical computing tools like(R/Python) will be used. Before conducting any analysis, data will be cleaned and prepared. This will include checking for missing values, removing duplicates, and transforming variables if necessary.

Data analysis will include:-

1. **Descriptive statistics:**
    - **This will be calculated to summarize the survey data, such as measures of central tendency, measures of variability, and frequency distributions.**
    - **Descriptive analysis provides a snapshot of the data and helps to identify patterns and trends.**

2. **Inferential statistics:(such as -> chi-square and t-test)**
    - **This involves using statistical tests to make inferences about the population based on sample data.**

3. **Correlation analysis:**
    - **This involves determining the degree and direction of the relationship between variables.**
    - **Correlation analysis can help identify patterns and relationships in the data, but it does not establish causation.**

4. **Regression analysis:**
    - **Which is used to predict one variable based on another variable or set of variables.**

    note:

    Correlation analysis and regression analysis are often used together to better understand the relationships between variables.

- **For example:-**

-> Correlation analysis can be used to identify variables that are significantly correlated with the response variable, which can then be included in a regression analysis to predict the value of the response variable.
-> Alternatively, regression analysis can be used to determine which variables have a significant impact on the response variable.
-> Correlation analysis can be used to examine the strength and direction of the relationships between those variables.

5. Data visualization:

- **This involves creating charts, graphs, or other visual representations of the data to help identify patterns and trends.**
- **Visualizing data can make it easier to communicate findings to stakeholders and decision-makers.**

6. Reporting the results:

- **This involves presenting findings in a clear and concise manner, including any limitations or caveats to the analysis.**
- **The report should also include recommendations based on findings and implications for future actions.**

Timeline:

Week 1: Planning:-

- **During this phase, research goals and objectives will be defined. Furthermore, we will identify the target population as well as developing the survey design.**

Week 2-3: Preparing for Questionnaire & Sampling and Pre-tests:-

- **In this phase, we will finalize the survey questions and formats, choose the survey administration method (online, mail, or in-person), defining the sample (The group of students whom the survey is intended to represent) and pretesting(identify any problem with the survey questions, format or delivery before it is given to the larger group).**

Week 4: collect the data:-

- **In this phase, survey responses will be collected. Moreover, we will do pre-analysis in order to obtain a clean, well framed and organized survey dataset.**

Week 5: Expectation vs Real:-

- **In this phase, we will know the total number of answered, non answered surveys and the number of surveys that will be taken in consideration and decide if the survey meets our expectations or not and whether it is valid to do analysis on.**

Week 6: Data Analysis:-

- **In this phase, identifying trends and patterns is important. Moreover, We will develop a report of survey findings as well.**

Week 7: Reporting and Presentation:-

- **During this phase, we will summarize survey findings in a report, develop a presentation of survey findings and present the report with the findings that we have reached.**

So, we expected that the survey timeframe would range from 6 to 7 weeks.

Budget:

The purpose of this activity is to think about the costs of implementing a survey and the impact of alternative survey procedures on the costs of a survey, data collection and the quality of data derived from that design.

For example:-

- **There are different fixed costs associated with a telephone survey, which requires the purchase and the use of physical telephones.**
- **An internet survey cost is totally different from telephone survey as it does not require that infrastructure.**

Fixed Costs:-

Estimate total fixed costs for the project:

- **Transportation allowance**
- **Sampling plan**
- **Data collection materials (computers, handhelds, telephones)**
- **Questionnaire preparation**
- **Data cleaning and analysis**
- **Other fixed costs**

Incidental Costs:-

- **Itemize and estimate the cost of any incidental costs (e.g., contingencies for lost or stolen materials).**
- **Incidental costs may not actually be realized, but you need to take it into consideration in the budget in case they occur.**

Sampling:-

- **Sampling refers to the process of selecting participants to take part in the survey.**
- **The cost of sampling may vary depending on sample size and sampling methods used.**
- **If you have a small sample size and can reach your participants through existing networks, social media or emails, sampling may be free.**
- **However, if you need to reach a large sample population or if you need to use a specific target audience, you may need to use paid sampling services, which can be expensive.**
- **This may include the cost of purchasing email lists, advertising through social media, or partnering with market research firms.**

Data Collection:-

- **The cost of data collection will depend on the mode of data collection.**
- **Online surveys are typically less expensive than telephone or face-to-face surveys.**
- **If you plan to conduct the survey online, you may need to consider the cost of survey softwares or platforms that charge fees.**
- **If you plan to conduct the survey through other modes such as phone or mail, you may need to consider the cost of sending emails or even talking on the phone.**

Data Analysis:-

- **The cost of data analysis will depend on the complexity of the data collected and the software used for analysis.**

Overhead Expenses:-

- **Overhead expenses refer to the general costs associated with conducting the survey such as office supplies, software, and equipment rental.**
- **You may need to take in consideration the cost of purchasing supplies, renting equipment, or using software that charges a fee.**

Overall, the cost of conducting a survey can vary depending on the scope and complexity of the project. It's important to carefully plan and budget for each stage of the survey to ensure that you have sufficient resources to complete the project.

Ethics and Confidentiality:-

1. Confidentiality:

- **The survey will be conducted anonymously and responses collected will be kept strictly confidential. Participant's identities will not be revealed and their personal information will not be shared with any third parties.**

2. Informed Consent:

- **The survey will include a statement that explains the purpose of the survey and informs participants about the use of data collected. Participants will be asked to provide their consent to participate in the survey before answering the questions.**

3. Data Protection:

- **All data collected from the survey will be stored securely and protected from unauthorized access. Data will be used solely for research purposes and any published results will be presented in an aggregated format to ensure individual responses cannot be identified.**

4. Ethics:

- **The survey will comply with ethical guidelines for research involving human subjects. The research will be conducted in a manner that respects participant's rights and dignity.**

5. Voluntary Participation:

- **Participation in the survey will be entirely voluntary and participants will have the option to withdraw from the survey at any time.**

6.Contact Information:

- **Participants will be provided with contact information of the research team in case they have any questions or concerns about the survey.**

By including a section on ethics and confidentiality, we can ensure that participant's privacy and rights are protected throughout the survey. This will help to increase the response rate and ensure that the data collected is of high quality.

Conclusion:-

'Food preferences' survey is a vital tool for collecting data on people's food choices and preferences. Data collected will provide insights into people's dietary habits and help in developing effective marketing strategies and menu items. By understanding food preferences, food service providers can improve customer satisfaction, promote healthy eating habits and reduce food waste.